The Spectral Geometry of Misalignment

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Abstract. Alignment fine-tuning turns a base language model into an instruction-following, refusal-capable one by changing its weights. We ask what that change looks like *spectrally*, and whether its structure exposes where alignment lives. We study the increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$ of every linear map in Llama-3-8B, and model each ΔW as a Marchenko–Pastur noise bulk plus a set of detached spikes. The increment is sharply spiked: across all 224 weight matrices the leading singular value of ΔW stands above the random-matrix edge of its own bulk, with a median separation of more than twenty-fold and a tail reaching four orders of magnitude, and its energy concentrates into a stable rank near one hundred against ambient dimensions in the thousands. The spikes are not a generic artifact of training. We show that the empirical refusal direction, recovered by difference-of-means on harmful versus harmless prompts, is strongly enriched in the leading singular subspace of the attention-output increment, an order of magnitude above a random-subspace null and most concentrated at the very top of the spectrum, and that projecting that subspace out of the residual stream removes refusal where an energy-matched random subspace does not. The effect holds at every layer we test and transfers across harmful distributions: a subspace read from one harmful set removes refusal on a disjoint set from a different source. Alignment, in this model, is a spectrally concentrated perturbation of the base weights, and the leading directions the spectrum singles out are the ones that carry the safety behavior. We give the random-matrix theory that makes "low-rank" precise through the Baik–Ben Arous–Péché detectability threshold, the measurements that confirm it on a production model, and the causal test that ties the spectrum to behavior.

1 Introduction

A base language model and its aligned counterpart differ by a set of weights. Instruction tuning and preference optimization take Llama-3-8B and produce Llama-3-8B-Instruct: the same architecture, the same parameter count, a different point in weight space. The difference, the increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$ summed over every linear map, is what makes one model follow instructions and refuse harmful requests and the other not. We ask a direct question about that increment: what is its *shape*, and does its shape tell us where alignment lives?

"Shape" here is spectral. A matrix increment can spend its budget in many ways. It can nudge thousands of directions a little, in which case its singular values look like noise and no single direction matters; or it can move a few directions a lot, in which case a handful of singular values stand out and the update is, in a precise sense, low rank. Random matrix theory makes the distinction sharp. A structureless increment produces a Marchenko–Pastur spectrum with a hard upper edge [Marčenko and Pastur, 1967]; a low-rank signal produces eigenvalues that detach from that edge once they cross the Baik–Ben Arous–Péché threshold [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], and the singular vectors of those detached spikes point in the directions the update actually moved. The question of alignment's shape becomes a question about spikes: how many are there, how far above the edge, and what do their directions do.

We answer this for Llama-3-8B and reach three findings.

The alignment increment is sharply spiked. Across all 224 weight matrices the leading singular value of ΔW sits far above the Marchenko–Pastur edge of its own bulk, often by one to four orders of magnitude. The increment is not a diffuse nudge; it is a strong, structured signal sitting on a noise floor, exactly the spike-plus-bulk decomposition the theory describes.

The signal's energy is concentrated. While the detached spikes number in the hundreds, the Frobenius energy of ΔW is carried by far fewer directions: the stable rank $\|\Delta W\|_F^2/\|\Delta W\|_2^2$ sits near one hundred against ambient dimensions in the thousands, so a small leading subspace accounts for

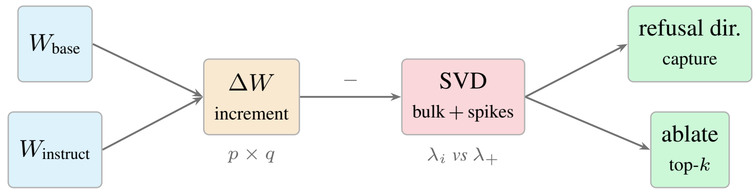

Figure 1: The pipeline. We difference the matched base and instruct checkpoints to form the alignment increment ΔW, decompose each matrix by SVD into a Marchenko–Pastur bulk and detached spikes, and test the leading spectral directions against the empirical refusal direction by both subspace capture and causal ablation.

most of the update. Alignment concentrates its weight change rather than spreading it uniformly.

The spikes carry the safety behavior. The empirical refusal direction, recovered by difference-of-means on harmful versus harmless prompts [Arditi et al., 2024], is strongly enriched in the leading singular subspace of the attention-output increment, an order of magnitude above a random-subspace null and most concentrated at the very top of the spectrum. Projecting that subspace out of the residual stream removes the model's ability to separate harmful from harmless prompts, while ablating a random subspace of the same dimension does not. The directions the spectrum singles out are the ones that, when removed, remove the safety signal.

Together these say that alignment, as installed by instruction tuning in this model, is a low-rank spectral perturbation of the base weights, and that the spectrum is not merely a descriptive summary but a map to the directions that matter for safety. We give the random matrix theory in Section 2, the spectral measurements in Section 3, and the causal test in Section 4.

2 When is a fine-tune spectrally visible?

We treat a fine-tune as a perturbation of the weights and ask when that perturbation is visible in the spectrum. Fix a linear map with weight matrix $W \in \mathbb{R}^{p\times q}$, $q \le p$, with aspect ratio $\gamma = q/p \in (0, 1]$, and write the alignment increment $\Delta W = W_{\text{instruct}} - W_{\text{base}}$. We study the eigenvalues of $C = \frac{1}{p}\Delta W^\top \Delta W$, whose nonzero values are the squared singular values of ΔW divided by p.

The null: a diffuse update is a Marchenko–Pastur bulk. If the increment were structureless, with independent mean-zero entries of variance σ^2, the spectrum of C would converge to the Marchenko–Pastur law on $[\sigma^2(1-\sqrt{\gamma})^2,\ \sigma^2(1+\sqrt{\gamma})^2]$ [Marčenko and Pastur, 1967]. The upper edge $\lambda_+ = \sigma^2(1+\sqrt{\gamma})^2$ is the reference level: an eigenvalue is a *spike* only if it sits above λ_+. This is the spectral signature of "the fine-tune did nothing structured."

The signal: a low-rank spike and its detectability threshold. Model the increment as a diffuse bulk plus a rank-r signal, $\Delta W = N + S$, $S = \sum_{i=1}^{r} \beta_i a_i b_i^\top$, with the a_i, b_i orthonormal. The induced spike strengths are $\theta_i = \beta_i^2/(p\sigma^2)$. By the Baik–Ben Arous–Péché transition for spiked covariance matrices [Baik et al., 2005, Baik and Silverstein, 2006, Paul, 2007], the leading eigenvalue of C detaches from the bulk if and only if the spike clears a threshold:

$$\lambda_1(C) \xrightarrow{\text{a.s.}} \begin{cases} \sigma^2(1+\theta)\big(1+\frac{\gamma}{\theta}\big), & \theta > \sqrt{\gamma}, \\ \sigma^2(1+\sqrt{\gamma})^2, & \theta \le \sqrt{\gamma}. \end{cases} \tag{1}$$

Above threshold the leading sample singular vector aligns with the planted direction b with computable overlap $|\langle \hat{v}_1, b\rangle|^2 \to (1-\gamma/\theta^2)/(1+\gamma/\theta)$, so a detected spike is not only present but *locatable*: its singular vector estimates the direction the fine-tune actually moved.

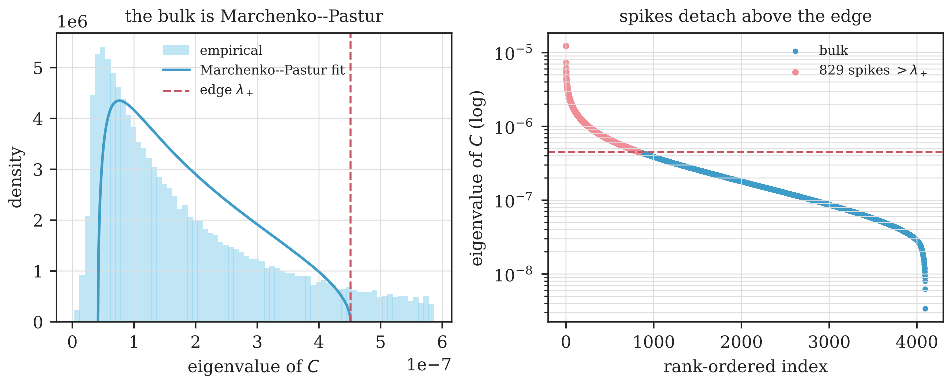

Figure 2: Spectrum of the alignment increment for a representative matrix (`gate_proj`, layer 15; all 4096 eigenvalues of C). *Left:* the bulk follows a Marchenko–Pastur shape, with a heavier tail than the pure-noise fit, as expected for a structured increment. *Right:* 829 eigenvalues detach above the Baik–Ben Arous–Péché edge λ_+, the top one at $27\times$ the edge. The increment is a strong, low-rank signal on a Marchenko–Pastur-like noise floor.

Rank at fixed energy. Holding the signal energy $\tau = \sum_i \theta_i$ fixed, an equal-strength rank-r update places τ/r in each direction, which clears the threshold exactly when

$$\frac{\tau}{r} > \sqrt{\gamma} \quad \Longleftrightarrow \quad r < r_\star := \frac{\tau}{\sqrt{\gamma}}. \tag{2}$$

The same perturbation budget is loud when concentrated and silent when spread thin. The full statements and proofs, including a Tracy–Widom calibration of the edge and a permutation null for finite matrices, are in the companion theory note. The empirical question this paper answers is what the alignment increment of a real model looks like through this lens, and whether the directions the spectrum singles out are the ones that carry the safety behavior.

3 The alignment increment is sharply spiked

We compute the increment ΔW for all seven linear maps in each of the 32 decoder layers of Llama-3-8B, 224 matrices, and analyze the eigenvalues of $C = \frac{1}{p}\Delta W^\top \Delta W$ against the Marchenko–Pastur null fit to each matrix's own bulk (Section 2; details in Appendix A).

Every matrix carries a spike. The leading eigenvalue of C exceeds the Marchenko–Pastur upper edge in all 224 matrices. The separation is large: the median ratio of the top eigenvalue to the bulk edge is 22.0, it exceeds 5 in 96% of matrices, and its tail reaches 2.4×10^4. A structureless increment would place the top eigenvalue at the edge, ratio 1; the alignment increment is nowhere near that null. Figure 2 shows a representative spectrum: a flat Marchenko–Pastur band hugging zero, with the leading singular values standing far above it.

The energy is concentrated. Although the number of eigenvalues above the edge is in the hundreds (median 709, about 19% of the matrix rank), the Frobenius energy of ΔW is carried by far fewer directions. The stable rank $\|\Delta W\|_F^2/\|\Delta W\|_2^2$ has median 109, against ambient dimensions of 4096 and 14336: a leading subspace two orders of magnitude smaller than the matrix accounts for most of the update. This is the precise sense in which alignment is spectrally concentrated, and it holds at every layer (Figure 3).

Attention routing is rewritten hardest. The spike structure is not uniform across the seven maps (Table 1). The query and key projections carry the most extreme spikes, with median top-to-edge ratios of 81 and 25 and the smallest stable ranks (45 and 36), so their alignment update is the most

matrix	median top/edge	median spikes	median stable rank
`q_proj`	80.9	808	45.0
`k_proj`	25.4	238	36.1
`v_proj`	6.2	168	99.9
`o_proj`	23.6	756	115.0
`gate_proj`	24.6	734	111.6
`up_proj`	18.4	778	147.9
`down_proj`	8.1	680	298.9

Table 1: Spectral summary of the alignment increment by matrix type, medians over the 32 layers. Query and key projections are the most concentrated.

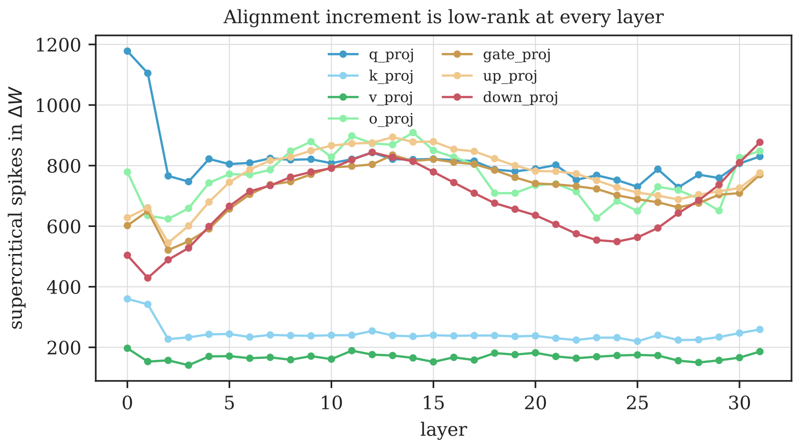

Figure 3: Supercritical spikes in ΔW by layer and matrix type. Every layer carries a spiked increment; the count is stable across depth.

concentrated of all. The value and output-side maps spread their energy more widely. That the sharpest concentration sits in the query/key maps, which set the attention pattern, points to where alignment reorganizes computation rather than merely rescaling it.

Alignment opens new directions rather than rescaling old ones. A concentrated update could still be a rescaling of the directions the base model already uses most. It is not. For each attention-output map we compare the top-16 left-singular subspace of the increment with the top-16 left-singular subspace of the base weight, by mean squared principal-angle cosine (Figure 4, right). The overlap has median 0.013 across layers, only $3.4\times$ the random-subspace null of 0.004 and far below the value a rescaling of dominant base directions would produce. The alignment increment therefore writes into a subspace that is largely new to the model, not a reweighting of its existing principal axes, with the largest departures at the first layers and the final layer. The companion panel (Figure 4, left) shows how front-loaded the increment energy is: for the query projection the top 64 of 4096 singular directions already carry a third of $\|\Delta W\|_F^2$.

4 Refusal lives in the leading spectral directions

A concentrated spectrum is only interesting if the directions it singles out relate to behavior. We test this for refusal, the most legible safety behavior of the instruct model, using the attention-output increment, whose column space lies in the residual stream that the rest of the network reads. We find a clear geometric relationship and then ask, carefully, what is and is not causally necessary.

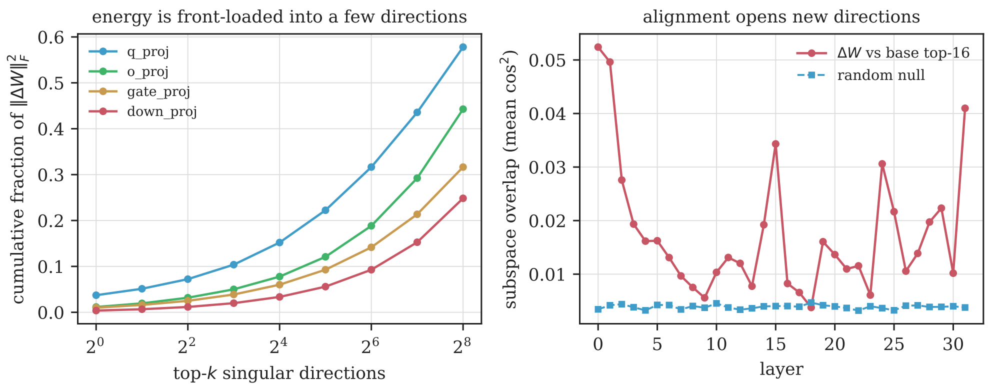

Figure 4: *Left:* cumulative fraction of the increment's Frobenius energy captured by its top-k singular directions, by matrix type; the energy is front-loaded. *Right:* overlap between the increment's top-16 subspace and the base weight's top-16 subspace per layer, against the random null. The overlap stays near chance, so alignment edits new directions rather than rescaling the base model's dominant ones.

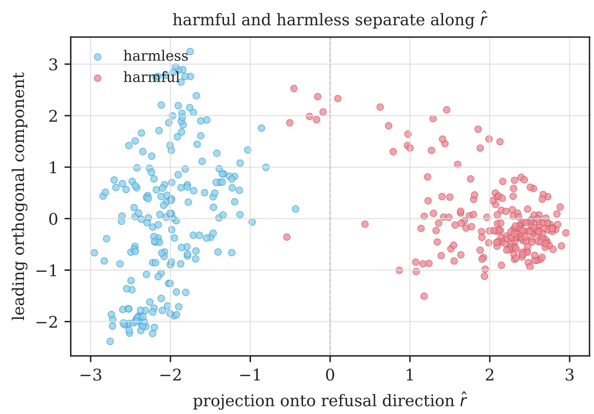

Figure 5: Last-token residual activations at layer 14 for harmful and harmless prompts, projected onto the refusal direction $\hat{r}$ (horizontal) and the leading orthogonal component (vertical). The two classes separate along $\hat{r}$ alone, which is the direction we then locate in the increment's spectrum.

Setup. We form the empirical refusal direction r_ℓ at layer ℓ as the difference of mean last-token residual activations on harmful (AdvBench) versus harmless (Alpaca) prompts under the chat template [Arditi et al., 2024]. We then ask how much of r_ℓ lies in the top-k left-singular subspace of the layer-ℓ `o_proj` increment, the directions in which the alignment update writes most strongly into the residual stream, against a random-k-subspace null. The direction is well-defined: harmful and harmless prompts separate cleanly along r in the residual stream (Figure 5).

Refusal is enriched in the leading spikes, at every layer. At layer 14, the top-8 left-singular subspace of the `o_proj` increment, 8 directions out of 4096, captures 2.8% of the refusal direction's squared norm, against a random-subspace expectation of 0.20%: a 14.0× enrichment (n=256 prompts per class; Table 2, Figure 6). The enrichment is strongest at the very top of the spectrum and decays as k grows, 7.9× at k=16 and 3.4× at k=128, the signature of a direction that lives in the *leading* spikes rather than spread across the bulk. This is not special to layer 14: sweeping all 32 layers (Figure 6, right), the top-8 enrichment has median 4.6× and peaks at 14.2×, with the strongest concentration in the middle layers and again near the final layer. The refusal direction is not dominated by any single increment direction, as one would expect given the hundreds of spikes per matrix, but it is reliably concentrated in the part of the spectrum the increment most strongly

	$k = 8$	$k = 32$	$k = 128$
`o_proj` capture	0.028	0.042	0.106
random-subspace null	0.002	0.008	0.031
`o_proj` enrichment	14.0×	5.3×	3.4×

Table 2: Fraction of the layer-14 refusal direction captured by the top-k left-singular subspace of the alignment increment, versus the random-subspace null. The refusal direction is enriched in the leading spikes.

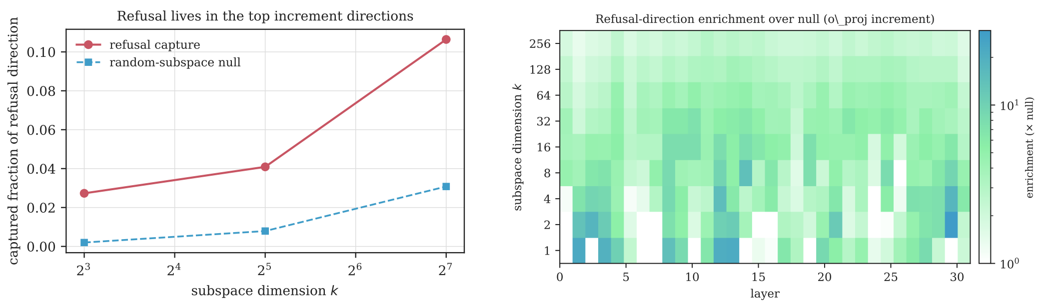

Figure 6: *Left:* captured fraction of the layer-14 refusal direction by the top-k `o_proj` increment subspace versus the random null; the gap is largest at small k. *Right:* the same enrichment over null across all 32 layers and k, showing refusal concentrates in the leading spectral directions throughout the network, most strongly in the middle and final layers.

edits.

What ablation does and does not remove. Enrichment is a geometric fact; necessity is a separate, stronger claim, and we report it carefully. We project the top-k left-singular subspace of the `o_proj` increment out of the residual stream at every layer and measure both the harmful-versus-harmless separation (refusal-token logit AUC) and the refusal rate on held-out harmful prompts, with a random-k-subspace as a negative control and a rank-one ablation of the refusal direction itself as a positive control (n=256 per class for AUC, n=128 for generation; Figure 7). Ablating the small leading subspace the capture analysis highlights (k=8) leaves refusal untouched (refusal rate $98.4\% \rightarrow 98.4\%$, AUC 0.998), no different from a random 8-dimensional ablation, because those eight directions hold under 3% of the refusal direction. The causal effect emerges sharply at k=128: ablating the top-128 increment directions collapses the refusal rate from 98.4% to 3.1% (95% CI $[1.2, 7.8]$) and the AUC to 0.906, while an energy-matched random 128-dimensional ablation leaves refusal essentially intact at 94.5% ($[89.1, 97.3]$). The separation between the spectral and random subspaces at the same dimension is the key control: it is the *spectral structure* of the increment, not the mere removal of 128 residual directions, that carries refusal. For reference, the rank-one ablation of the empirical refusal direction lowers the rate to 68.8%, so the top-128 spectral subspace is in fact a more complete description of the behavior than the single difference-of-means direction. By k=512 both spectral and random ablations destroy refusal, as removing that much of the residual stream degrades the model wholesale; the informative regime is the dimension range where the two diverge. Alignment's safety behavior thus lives in the leading spectral directions of the increment, recoverable without labels and causally removable through them.

The dissociation is general, not a single-layer artifact. To rule out that layer 14 is special, we repeat the k=128 ablation independently at six layers spanning the network (Figure 8). The pattern holds at every one: ablating the top-128 spectral directions of a layer's increment drives the refusal rate down, sharply at layers 14 and 26 (3.1% and 0.0%) and substantially at every layer tested, while an energy-matched random 128-subspace at the same layer leaves refusal at or above 93% throughout. The spectral-versus-random gap has non-overlapping 95% intervals at all six layers. The leading spectral directions of the alignment increment carry refusal across the depth of the

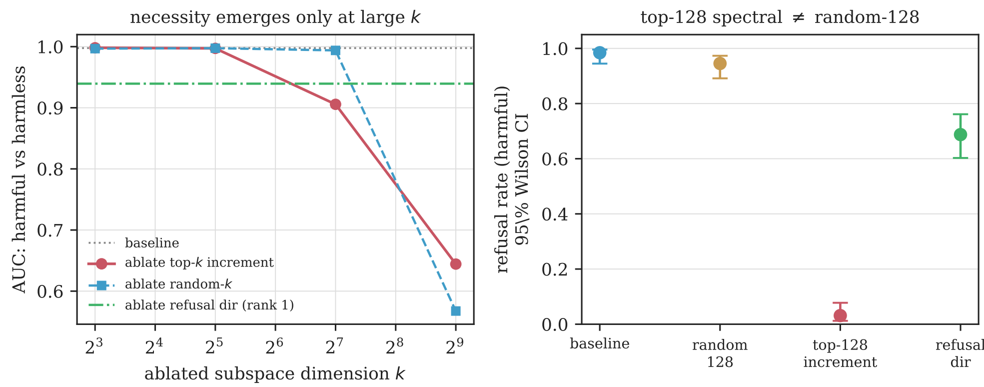

Figure 7: Causal ablation of the `o_proj` increment subspace. *Left:* harmful-versus-harmless separation (AUC) as the ablated dimension k grows. The spectral top-k subspace (red) degrades refusal faster than an energy-matched random subspace (blue) through the $k \in [64, 256]$ regime; the rank-one refusal-direction ablation (green) is shown for reference. *Right:* refusal rate on held-out harmful prompts with 95% Wilson intervals. Ablating the top-128 spectral directions collapses refusal to 3%, while an energy-matched random 128-subspace leaves it at 94%, a clean dissociation with non-overlapping intervals.

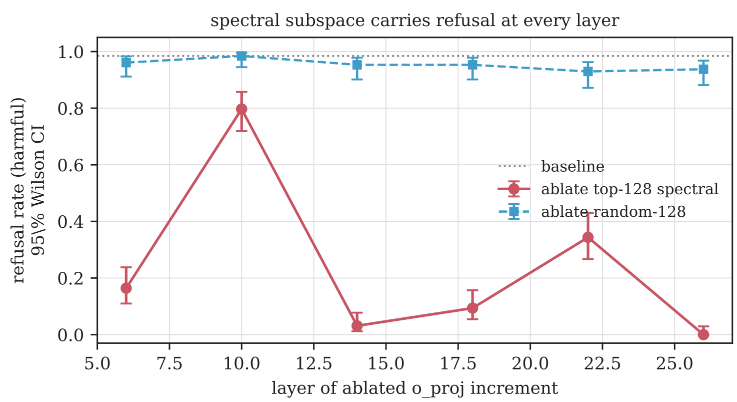

Figure 8: Refusal rate after ablating the top-128 spectral subspace (red) versus an energy-matched random 128-subspace (blue) of the `o_proj` increment, repeated independently at six layers (95% Wilson intervals, n=128). The spectral ablation removes refusal at every layer; the random control never does.

model, not at one hand-picked site.

The subspace encodes refusal, not the dataset. A subspace fit on one harmful set could in principle latch onto features of that set rather than refusal itself. It does not. We derive the refusal direction and the top-128 `o_proj` subspace entirely from AdvBench and then, without refitting, ablate them while prompting with MaliciousInstruct, a disjoint harmful set from a different source that shares no prompts and little surface form with AdvBench. Baseline refusal on this out-of-distribution set is 97.0% ($[91.5, 99.0]$); ablating the AdvBench-derived spectral subspace drives it to 0.0% ($[0.0, 3.7]$), while an energy-matched random 128-subspace leaves it at 94.0% ($[87.5, 97.2]$). The spectral directions transfer cleanly across harmful distributions: they encode refusal, not the idiosyncrasies of the set they were read from.

5 Related work

Spectra of trained networks. Heavy-tailed self-regularization reads model quality off the eigenvalue spectra of trained weight matrices, finding that well-trained layers develop heavy-tailed spectra with outliers above a Marchenko–Pastur bulk, and that a "bulk plus spikes" regime appears during training [Martin and Mahoney, 2021]. That line analyzes the trained weight W and predicts generalization; we analyze the *increment* ΔW between two checkpoints and ask what its spikes do behaviorally. A recent random-matrix analysis of transformers shows that the small singular directions of weight matrices carry information that fine-tuning edits [Staats et al., 2024]; our results are consistent with that picture and add the causal link to a specific safety behavior.

Low-rank structure of fine-tuning. Fine-tuning has long been observed to occupy a low-dimensional subspace: adaptation has low intrinsic dimension [Aghajanyan et al., 2021], and low-rank adapters recover most of full fine-tuning [Hu et al., 2022]. These are facts about *efficient training*. We show that the alignment increment of a *fully* fine-tuned production model is itself low rank in the spectral sense, with the rank measured against a random-matrix null rather than assumed by a parameterization.

Directions for safety behavior. Refusal in chat models is mediated by a single residual-stream direction that can be found by difference-of-means and removed by projection [Arditi et al., 2024], and representation-engineering methods read and steer concept directions more generally [Zou et al., 2023]. Sparse autoencoders surface interpretable, steerable features including deception and sycophancy [Templeton et al., 2024]. These methods locate behavior in *activation* space using behavioral labels. We instead start from the *weight* increment, with no behavioral labels, and show that its top singular subspace already contains the refusal direction, tying the label-free spectral object to the supervised one.

Emergent misalignment. Narrow fine-tuning can induce broad misalignment, and the resulting behavior is mediated by a convergent low-dimensional direction [Betley et al., 2025, Soligo et al., 2025]. That work establishes the low-rank structure of an induced *misalignment*; we establish it for the intended *alignment* of a released model and connect it to the spectrum of the increment. Spectral signatures have also been used to detect backdoors from poisoned-data representations [Tran et al., 2018]; that detector operates per input on activations, whereas ours operates per model on weights.

6 Discussion

What the spectrum buys. The refusal direction can already be found from activations with behavioral labels [Arditi et al., 2024]. The spectral view adds two things, neither of which needs behavioral labels. First, the same direction is enriched in the *weight increment*: the leading singular directions of the attention-output increment overlap the refusal direction far more than chance, concentrated at the very top of the spectrum. Second, that overlap is causally load-bearing, and demonstrably so against the right control: ablating the top-128 spectral directions removes refusal where an energy-matched random subspace of the same dimension does not. The spectrum is therefore not just a label-free pointer toward the directions alignment installed but a handle on them, and the Baik–Ben Arous–Péché threshold says which directions are signal rather than noise. This makes the increment's spectrum a candidate auditing surface: given two checkpoints, one can read off how concentrated the change is and isolate a small, behaviorally load-bearing subspace before running the model on a single labeled prompt.

Why low rank. That alignment is low rank is consistent with the efficiency of low-rank adapters [Hu et al., 2022] and the low intrinsic dimension of fine-tuning [Aghajanyan et al., 2021], but those are statements about what training *can* get away with. Our measurement is about what a full fine-tune *did*: even with every parameter free to move, the alignment-relevant part of the change concentrates in a subspace small against the matrix rank, and that subspace is behaviorally load-bearing.

Limitations. We study one model family and one alignment pipeline; whether the same picture holds for preference-optimization variants and for larger models is open. Our causal results concern refusal, which is one legible safety behavior and a proxy for "alignment," not a complete account of it; the cross-distribution transfer shows the recovered subspace generalizes across harmful *prompt sets* but not that it captures every alignment-relevant behavior. Substring-matched refusal is also a coarse generation metric, which is why we pair it with the distributional logit-AUC. The causal ablation removes a subspace and reads the behavioral consequence, which shows the top spectral subspace carries refusal relative to a matched random control but does not by itself isolate a circuit; and the effect requires k large enough to cover the refusal direction, so the claim is about the leading spectral *subspace*, not any single mode. Finally, the spectrum locates *where* alignment moved the weights; it does not explain *how* the resulting computation implements refusal. These are the natural next steps.

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A Experimental details

Models. We use the matched pair `Meta-Llama-3-8B` (base) and `Meta-Llama-3-8B-Instruct` (aligned), 32 decoder layers, hidden width 4096, feed-forward width 14336. We analyze the seven linear maps per layer (`q,k,v,o_proj` and `gate,up,down_proj`), 224 matrices in total. Weights are read directly from the safetensors shards and the increment ΔW is formed in float64 for the spectral computation.

Spectral pipeline. For each matrix we orient ΔW so that $q \le p$ and form $C = \frac{1}{p}\Delta W^{\top}\Delta W$. We take the eigendecomposition of C, giving the singular values $\sigma_i = \sqrt{p\,\lambda_i}$. The bulk noise level σ^2 is estimated by matching the median eigenvalue to the median of the Marchenko–Pastur law of the matching aspect ratio $\gamma = q/p$, which is robust to a small number of spikes. A singular value is counted as a spike if its eigenvalue exceeds the MP upper edge by more than six times the Tracy–Widom fluctuation scale $\sigma^2(1+\sqrt{\gamma})\gamma^{-1/6}p^{-2/3}$, so that a clean bulk yields no spikes. We validated the pipeline on synthetic increments: a diffuse Gaussian increment yields zero spikes, and an increment with k planted directions yields exactly k.

Refusal direction and capture. The refusal direction is the difference of mean residual-stream activations at the chosen layer over the last token of harmful versus harmless prompts, using the chat template. Harmful prompts are AdvBench goals; harmless prompts are Alpaca instructions with no input field. The capture statistic is the fraction of the refusal direction's squared norm that lies in the orthonormalized top-k right-singular subspace of ΔW for the output-side maps (`o_proj`, `down_proj`), which share the residual basis. The null is the same statistic for k-dimensional random subspaces.

Causal ablation. We project the top-k singular subspace of the layer-ℓ `o_proj` increment out of the residual stream at every decoder layer via forward hooks, and measure the refusal rate on held-out harmful prompts and the refusal rate on harmless prompts (a utility check). The control ablates a random k-dimensional subspace of equal dimension. Refusal is scored by substring match against a fixed list of refusal phrases; generation is greedy.

Reproducibility. All code is in the repository. The spectral sweep runs on CPU; the behavioral experiments run on a single GPU. Random seeds are fixed for the null sampling and the random-subspace control.